

Sri Lanka Institute of Information Technology



Lab Sheet 02 2026-Y2-S1-MLB-B4G1-06 Web Based Video Browsing System

Student ID	Name
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IT25101767	Withana T. T.
IT25103673	Wickramathunga P. P.
IT25100859	Adikari A. M. T. D.

2026 - Semester 01

Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

1. Group Member Details & Assigned Scrum

Student Name	IT Number	SCRUM Role	Assigned Major Function
De Silva L. C. A.	IT25101841	Scrum Master	Announcement & Event System
Munasinghe M. H. V. S.	IT25102695	Developer	Ticketing System
Bandara W. G. M. D.	IT25103702	Developer	Advertisement Promotions
Withana T. T.	IT25101767	Developer	Course & Curriculum Management
Wickramathunga P. P.	IT25103673	Developer	Video Management
Adikari A. M. T. D.	IT25100859	Product Owner	Quiz & Assessment Subsystem

2. Identified Personas (6 Personas)

1. **Persona 1: Student / Learner** – End-users who register accounts, enroll in courses, watch educational videos, complete timed quizzes, and submit support tickets.
2. **Persona 2: Teacher / Instructor** – Educators responsible for uploading video lessons, creating course modules, configuring timed quizzes, and publishing academic announcements.
3. **Persona 3: Content Moderator** – Platform staff who inspect reported videos or comments, enforce community standards, quarantine violating media, and manage banned keyword lists.
4. **Persona 4: Marketing Executive** – Promotional leads responsible for creating homepage ad banners, highlighting featured courses, and analyzing platform engagement metrics.
5. **Persona 5: Operations Manager** – Technical administrators managing user role permissions, tracking support ticket statuses, and auditing platform activity logs.
6. **Persona 6: Guest / Unregistered User** – Visitors who browse public course listings, search videos by categories, preview educational content, and register for new accounts.

3. User Stories

Persona 1: Student / Learner

- **US-01:** As a Student, I want to search educational videos by title, subject, or category tags, so that I can quickly find relevant study materials.
- **US-02:** As a Student, I want to take timed quizzes attached to course modules, so that I can evaluate my understanding and receive instant feedback.
- **US-03:** As a Student, I want to view my watch history and bookmarked lessons, so that I can resume learning where I left off.
- **US-04:** As a Student, I want to submit support tickets regarding technical or course issues, so that I can receive assistance from platform admins.

Persona 2: Teacher / Instructor

- **US-05:** As an Instructor, I want to upload educational videos with titles, descriptions, and category tags, so that I can deliver course content to enrolled students.
- **US-06:** As an Instructor, I want to create timed quizzes with custom question banks and passing scores, so that I can assess student learning progress.
- **US-07:** As an Instructor, I want to publish announcements and schedule upcoming academic webinars/events, so that students stay informed about deadlines and live sessions.
- **US-08:** As an Instructor, I want to track student enrollment and quiz completion results, so that I can measure course effectiveness.

Persona 3: Content Moderator / Platform Admin

- **US-09:** As a Content Moderator, I want to review flagged or reported inappropriate videos and comments, so that I can maintain a safe learning environment.
- **US-10:** As a Content Moderator, I want to quarantine violating content and issue formal warnings to rule-breaking users, so that platform guidelines are enforced.
- **US-11:** As an Admin, I want to manage video categories and banned keywords, so that content remains organized and search results are safe.
- **US-12:** As an Admin, I want to view system analytics on active users, total video views, and reported content, so that I can monitor overall platform health.

Persona 4: Marketing Executive

- **US-13:** As a Marketing Executive, I want to create and schedule promotional advertisement banners on the homepage, so that featured courses and special offers are highlighted.
- **US-14:** As a Marketing Executive, I want to edit active advertisement details, display dates, and promotional images, so that campaigns remain up to date.
- **US-15:** As a Marketing Executive, I want to feature trending videos and top instructor profiles on the landing page, so that user engagement is maximized.
- **US-16:** As a Marketing Executive, I want to view engagement reports including advertisement click counts and popular search keywords, so that I can optimize marketing campaigns.

Persona 5: Operations Manager

- **US-17:** As an Operations Manager, I want to view and manage all user accounts (Student, Instructor, Admin), so that I can ensure secure role-based access control.

- **US-18:** As an Operations Manager, I want to track and update support ticket statuses (Open, In Progress, Resolved), so that user technical issues are handled efficiently.
- **US-19:** As an Operations Manager, I want to configure platform security settings and automated system notifications, so that operations run with minimal downtime.
- **US-20:** As an Operations Manager, I want to audit system logs and activity records, so that data safety and operational compliance are maintained.

Persona 6: Guest / Unregistered User

- **US-21:** As a Guest user, I want to browse publicly available educational video listings and course previews, so that I can explore offered subjects.
- **US-22:** As a Guest user, I want to register a new account choosing a role (Student or Instructor), so that I can access full platform features.
- **US-23:** As a Guest user, I want to search videos by title and categories on the homepage, so that I can check if the platform offers content I need.
- **US-24:** As a Guest user, I want to access the platform support and contact page, so that I can submit inquiries before registering.

4. Product Backlog

Backlog ID	Persona	User Story Description	Priority
PBI-01	Guest	Register a new account with secure credentials and role selection	High
PBI-02	All Users	Log in and authenticate securely with role-based dashboard redirection	High
PBI-03	Guest / Student	Browse, search, and filter educational videos by title, category, or tags	High
PBI-04	Instructor	Upload new educational videos with metadata, thumbnails, and categories	High
PBI-05	Instructor	Create and organize course modules and structured learning paths	High
PBI-06	Student	Enroll in courses and view video lessons with playback controls	High
PBI-07	Instructor	Create timed quizzes with question banks, passing criteria, and feedback	High
PBI-08	Student	Complete timed quizzes, receive instant scores, and obtain certificates	High
PBI-09	Student / Admin	Report inappropriate videos or comments for administrative review	High

PBI-10	Moderator	Review reported content, quarantine items, and issue user warnings	High
PBI-11	Moderator	Manage banned keyword lists and enforce content compliance policies	Medium
PBI-12	Instructor	Publish platform announcements and schedule academic events/webinars	Medium
PBI-13	Student	Receive event notifications and view upcoming academic schedules	Medium
PBI-14	Student / Teacher	Submit technical support tickets with detailed problem descriptions	Medium
PBI-15	Admin	Review, update, and resolve user support tickets	Medium
PBI-16	Marketing Executive	Create, edit, and schedule promotional advertisement banners	Medium
PBI-17	Marketing Executive	Display promotional ads and featured courses based on active display periods	Medium
PBI-18	Student	View recently watched video history and resume learning sessions	Low
PBI-19	Student / Teacher	Update profile information, profile picture, and change security password	Low
PBI-20	Admin	Manage platform user accounts (activate, suspend, remove accounts)	High
PBI-21	Marketing Executive	Feature trending videos and top instructor profiles on homepage	Low
PBI-22	Admin	View platform analytics dashboard (views, uploads, report statistics)	Medium
PBI-23	Marketing Executive	View promotion performance reports and ad click analytics	Low
PBI-24	Operations Manager	Review operational audit logs and system performance metrics	Low

5. Sprint Planning

The above product backlog requirements have been divided into 4 development Sprints to cover the system implementation lifecycle from Week 6 to Week 13.

1st Sprint — Week 5 to Week 6

Selected User Stories (according to their priorities):

Sprint ID	User Story	Priority
SP1-01	As a Guest user, I want to register a new account choosing a role (Student or Instructor), so that I can access full platform features.	High

SP1-02	As an authorized user, I want to log in securely with role-based redirection so that I can access my dashboard.	High
SP1-03	As an Instructor, I want to upload educational videos with metadata, thumbnails, and categories, so that students can watch them.	High
SP1-04	As a Student / Guest, I want to search and filter educational videos by title, category, or tags, so that I can find study materials.	High

Sprint Backlog

Task Breakdown for User Story 1 and Time Estimation (User Registration Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-01.1	Design UI forms for Registration screen	5	De Silva L. C. A.
T-01.2	Create backend API endpoint for user registration	6	Withana T. T.
T-01.3	Implement password hashing and input validation	4	Wickramathunga P. P.
T-01.4	Setup database schema for Users and Roles	4	Adikari A. M. T. D.
T-01.5	Test registration flow and role assignment	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 2 and Time Estimation (Login & Security Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-02.1	Design Login UI with fields for username/password	4	De Silva L. C. A.
T-02.2	Implement JWT authentication mechanism	6	Withana T. T.
T-02.3	Implement role-based dashboard navigation logic	4	Bandara W. G. M. D.
T-02.4	Display authentication errors and success alerts	2	Wickramathunga P. P.
T-02.5	Test secure login and session management	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 3 and Time Estimation (Video Upload Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-03.1	Design responsive Video Upload interface	6	Wickramathunga P. P.

T-03.2	Create backend API for video file upload and storage	8	Wickramathunga P. P.
T-03.3	Validate video file size, format (MP4/MOV), and duration	4	Bandara W. G. M. D.
T-03.4	Store video metadata (title, category, tags) in database	5	Withana T. T.
T-03.5	Test video uploading and playback streaming	4	Munasinghe M. H. V. S.

Task Breakdown for User Story 4 and Time Estimation (Video Search & Discovery Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-04.1	Design search bar and category filter interface	5	De Silva L. C. A.
T-04.2	Implement backend search API (title, category, tags)	6	Withana T. T.
T-04.3	Create category browsing endpoints	4	Adikari A. M. T. D.
T-04.4	Test search functionality and filter performance	3	Munasinghe M. H. V. S.

2nd Sprint — Week 7 to Week 8

Selected User Stories (according to their priorities):

Sprint ID	User Story	Priority
SP2-01	As an Instructor, I want to create timed quizzes with question banks and pass marks, so that I can evaluate student progress.	High
SP2-02	As a Student, I want to take timed quizzes and receive instant scores and completion certificates, so that I can track my learning.	High
SP2-03	As a Student / User, I want to report inappropriate videos or comments, so that admins can review harmful content.	High
SP2-04	As a Content Moderator, I want to review flagged items, quarantine violating content, and issue warnings to users.	High

Sprint Backlog

Task Breakdown for User Story 1 and Time Estimation (Quiz & Question Bank Management)

Task ID	Task	Est. Hours	Assigned Member
T-05.1	Design Quiz Creation UI with question forms & timers	6	Adikari A. M. T. D.

T-05.2	Implement backend API for Quiz CRUD operations	7	Adikari A. M. T. D.
T-05.3	Build question bank database models and relationships	5	Withana T. T.
T-05.4	Test instructor quiz creation and editing flow	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 2 and Time Estimation (Quiz Submission & Certificate Generation)

Task ID	Task	Est. Hours	Assigned Member
T-06.1	Design student Quiz Taking UI with countdown timer	5	Adikari A. M. T. D.
T-06.2	Develop automated grading calculation engine	6	Withana T. T.
T-06.3	Build auto-generated completion certificate backend service	5	De Silva L. C. A.
T-06.4	Test quiz submission, timer expiration, and scoring	4	Munasinghe M. H. V. S.

Task Breakdown for User Story 3 and Time Estimation (Content Flagging & Reporting Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-07.1	Design report submission dialog on video watch page	4	Wickramathunga P. P.
T-07.2	Create backend endpoint to process user content reports	5	Bandara W. G. M. D.
T-07.3	Log report reasons, uploader info, and submission dates	3	Bandara W. G. M. D.
T-07.4	Test report submission and notification triggering	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 4 and Time Estimation (Content Moderation & Quarantine Vault)

Task ID	Task	Est. Hours	Assigned Member
T-08.1	Design Moderation Dashboard UI for admins	5	Wickramathunga P. P.
T-08.2	Build endpoints to approve, ignore, or quarantine reported content	6	Bandara W. G. M. D.
T-08.3	Implement banned keyword filtering algorithm	5	Withana T. T.

T-08.4	Implement warning issuance and user suspension logic	4	De Silva L. C. A.
T-08.5	Test complete moderation and quarantine workflow	3	Munasinghe M. H. V. S.

3rd Sprint — Week 9 to Week 10

Selected User Stories (according to their priorities):

Sprint ID	User Story	Priority
SP3-01	As an Instructor / Admin, I want to publish announcements and schedule webinars/events, so that students stay informed.	Medium
SP3-02	As a Student, I want to receive event notifications and view upcoming academic events, so that I don't miss live sessions.	Medium
SP3-03	As a User (Student/Teacher), I want to submit technical support tickets, so that I can resolve platform issues.	Medium
SP3-04	As an Admin, I want to review, respond to, and update ticket statuses (Open, In Progress, Resolved), so that user problems are addressed.	Medium

Sprint Backlog

Task Breakdown for User Story 1 and Time Estimation (Announcement & Event Management)

Task ID	Task	Est. Hours	Assigned Member
T-09.1	Design UI for publishing announcements and event schedules	5	De Silva L. C. A.
T-09.2	Build backend CRUD APIs for Announcements & Academic Events	7	De Silva L. C. A.
T-09.3	Implement scheduled auto-removal for expired events	4	Withana T. T.
T-09.4	Test announcement posting and editing features	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 2 and Time Estimation (Event Notification Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-10.1	Design student announcement board and calendar widget UI	5	De Silva L. C. A.
T-10.2	Create notification broadcast API for new events	5	Adikari A. M. T. D.

T-10.3	Test notification display across student dashboards	3	Munasinghe M. H. V. S.
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Task Breakdown for User Story 3 and Time Estimation (Support Ticket Submission)

Task ID	Task	Est. Hours	Assigned Member
T-11.1	Design Support Page and ticket submission UI	5	Munasinghe M. H. V. S.
T-11.2	Implement backend API to record support tickets	6	Munasinghe M. H. V. S.
T-11.3	Implement ticket status tracking view for users	4	Bandara W. G. M. D.
T-11.4	Test support ticket creation and user tracking flow	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 4 and Time Estimation (Admin Ticket Resolution Functionality)

Task ID	Task	Est. Hours	Assigned Member
T-12.1	Design Admin Ticket Queue UI with status filtering	5	Bandara W. G. M. D.
T-12.2	Create backend endpoint to update ticket status and add responses	6	Munasinghe M. H. V. S.
T-12.3	Trigger email/system alert when ticket status changes	4	Wickramathunga P. P.
T-12.4	Test admin ticket resolution and customer notification	3	Munasinghe M. H. V. S.

4th Sprint — Week 11 to Week 12

Selected User Stories (according to their priorities):

Sprint ID	User Story	Priority
SP4-01	As a Marketing Exec / Admin, I want to create and manage promotional ad banners, so that featured courses and offers are highlighted.	Medium
SP4-02	As a User, I want to see active promotional ads on the homepage based on display schedules, so that I learn about featured courses.	Medium
SP4-03	As an Admin, I want to view platform analytics (views, users, uploads, ticket stats), so that I can monitor overall system performance.	Medium

SP4-04	As an Admin / Operations Manager, I want to audit system activity logs and manage user accounts, so that access control is maintained.	High
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Sprint Backlog

Task Breakdown for User Story 1 and Time Estimation (Advertisement Banner Management)

Task ID	Task	Est. Hours	Assigned Member
T-13.1	Design Advertisement creation and banner manager UI	5	Bandara W. G. M. D.
T-13.2	Implement backend CRUD REST APIs for Advertisement entities	7	Bandara W. G. M. D.
T-13.3	Implement ad display duration and status toggle logic	4	Withana T. T.
T-13.4	Test advertisement banner creation, edit, and deletion	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 2 and Time Estimation (Dynamic Ad Display Engine)

Task ID	Task	Est. Hours	Assigned Member
T-14.1	Design homepage banner rendering widget	4	Bandara W. G. M. D.
T-14.2	Implement frontend dynamic ad fetcher based on active dates	4	Adikari A. M. T. D.
T-14.3	Test promotional ad scheduling and automatic expiry	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 3 and Time Estimation (Admin System Analytics Dashboard)

Task ID	Task	Est. Hours	Assigned Member
T-15.1	Design Admin System Analytics Dashboard UI with summary cards	6	Wickramathunga P. P.
T-15.2	Create database aggregation queries (views, uploads, reports)	7	Withana T. T.
T-15.3	Build data export/reporting endpoints	4	De Silva L. C. A.
T-15.4	Test analytics reporting accuracy and UI rendering	3	Munasinghe M. H. V. S.

Task Breakdown for User Story 4 and Time Estimation (User Account Management & Integration)

Task ID	Task	Est. Hours	Assigned Member
T-16.1	Design User Account Management table UI for admins	5	De Silva L. C. A.
T-16.2	Implement account suspend/activation REST endpoints	5	Withana T. T.
T-16.3	Perform full cross-module integration testing and bug fixing	8	All Members
T-16.4	Perform code documentation clean-up and viva prep	4	All Members